

Laboratory assignment

Component 2

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1 Problem Definition

Streamlining emergency services is important to give the correct treatment in time. In this context, the ambulance can be viewed as an abstract Multi-Agent System consisting of several interacting agents, including traffic agents, locators, teams of emergency medical responders, and most importantly, the ambulance coordinator. One ambulance coordinator, traffic agent, and tracking agent are assigned to each actual ambulance.

Each agent has its own assigned role who dictates its behaviour. When an emergency call is received, the Ambulance Coordinator initiates an auction to select the most suitable Ambulance Team based on their current location and capability. Once a team is assigned, the Tracing Agent provides feedback from previous movements, analyzing if past trips along that specific path were effective or if trajectories should be altered to ensure a timely arrival. As the vehicle moves, the Traffic Agent maintains a constant exchange with the national traffic center regarding road blockages or accidents. This agent immediately broadcasts these incident details to the whole squad and the Tracing Agent so the team can avoid unreachable locations. The Ambulance Team Agent, representing the real-world autovehicle, uses its internal GPS and trajectory modules to navigate the optimal path provided by these data exchanges. The process concludes with the Tracing Agent recording the mission's success or failure into a trace record to improve the dependability of future emergency responses.

2 High-Level MAS Specification

We are defining:

- $\mathcal{A} = \{A_1, A_2, \dots, A_m\}$ be the set of ambulances.
- $\mathcal{T} = \{T_1, T_2, \dots, T_m\}$ be the set of Ambulance Team Agents, where each T_i represents the vehicle A_i .
- $\mathcal{TA} = \{TA_1, TA_2, \dots, TA_m\}$ be the set of Traffic Agents, such that each TA_i is assigned to a specific ambulance A_i .
- $\mathcal{TR} = \{TR_1, TR_2, \dots, TR_m\}$ be the set of Tracing Agents, such that each TR_i is assigned to a specific ambulance A_i .
- AC be the **Ambulance Coordinator** agent in charge of the global auction and coordination process.

2.1 Inputs to the MAS System

- The number of agents: m (physical ambulances), where $|\mathcal{T}| = |\mathcal{TA}| = |\mathcal{TR}| = m$.
- A set of traffic incidents $\mathcal{I} = \{i_1, i_2, \dots, i_k\}$ provided by the national traffic center.
- Real-time Position: $\mathcal{G}_t : \mathcal{T} \rightarrow \mathbb{R}^2$ representing the GPS coordinates of each vehicle at time t .
- Historical Data: The set of trace records $\mathcal{R} = \{r_1, r_2, \dots, r_m\}$ containing success and failure records for previous paths.

2.2 Outputs from the MAS System

- Auction Results: The assignment $Log_t = (T_i, Patient_j)$ for a specific emergency call.
- Optimal Trajectories: The path sequence $\mathcal{P}_i = \{loc_1, loc_2, \dots, loc_n\}$ for ambulance i to avoid detected blockages.
- Updated Traces: The modified set of history records $\mathcal{R}_{t+1}$ after recording mission effectiveness.

2.3 Types of Agents

The agents are classified based on their architectural complexity and their specific roles within the emergency medical service:

Agent	Abstract Architecture	Concrete Architecture	Reason
Ambulance Coordinator	State-Based	Deliberative	Manages the auction method to choose the best crew for transport.
Ambulance Team	State-Based	Hybrid	Acts as a bidder in auctions and monitors routes using GPS modules.
Traffic Agent	Purely Reactive	Reactive	Broadcasts incident details immediately to the squad and tracing agent.
Tracing Agent	State-Based	Deliberative	Gathers movement data and provides feedback to improve dependability.

2.4 Specification of Agents

2.4.1 Traffic Agent (TA_i)

- **Inputs:** $Percepts_{TA} = \{Incident(loc, type)\}$ from the national traffic center.
- **Outputs:** $a = Alert(loc, blockage)$.
- **Task:** Monitor for road obstacles and broadcast events to the squad to pinpoint unreachable locations.

2.4.2 Tracing Agent (TR_i)

- **Inputs:** $Percepts_{TR} = \{MissionLog(T_i, loc, time)\}$.
- **Outputs:** Updated record $r_i \in \mathcal{R}$
- **Task:** Record mission effectiveness (1 for success, 0 for failure) and update historical trace files.

2.4.3 Ambulance Coordinator Agent (AC)

- **Inputs:** Emergency calls and bids received from the teams $\mathcal{T}$.
- **Outputs:** $Log_t = (T_i, Patient_j)$.
- **Task:** Execute auction-based selection for suitable crew dispatch.

2.4.4 Ambulance Team Agent (T_i)

Each ambulance team T_i operates using a behavior function:

$$f_{T_i} : Percepts_{T_i} \rightarrow Actions_{T_i} \quad (1)$$

$$Actions_{T_i} = \{bid(price, urgency), move(path), treat(patient), stay\}$$

2.5 Communications among A

The system uses **Publisher-Subscriber** for traffic alerts because it is fast and reaches everyone at once. It uses **Request-Response** for the Coordinator's auction and the Tracing Agent's feedback, ensuring that specific data is sent to the right agent at the right time.

Sender	Receiver	Message Type	Notation
Traffic TA_i	Squad $\mathcal{T}$	TrafficAlert	$a = Alert(loc, blockage)$
Coordinator AC	Team T_i	AuctionCall	$c = CFP(Patient_j, loc)$
Team T_i	Tracing TR_i	StatusUpdate	$s = Trace(trajjectory)$
Tracing TR_i	Team T_i	Feedback	$f = Feedback(\mathcal{F}_i)$

3 Agents' Role within the MAS

- **Traffic Agent (TA_i):** The information centre that identifies unreachable locations and updates the squad on road obstacles.
- **Tracing Agent (TR_i):** The data collector that analyzes previous movements to ensure optimal path selection.
- **Ambulance Team Agent (T_i):** The mobile responder that performs bidding and route monitoring using GPS technology.
- **Ambulance Coordinator (AC):** The central authority that selects the ambulance crew via an auction method.